BTS-II(R)-07-24-2824 Reg. No. [redacted]

A

B.Tech. Degree II Semester Regular Examination July 2024

EE/EC/CS/IT 23-200-0201B LINEAR ALGEBRA AND TRANSFORM TECHNIQUES

(2023 Scheme)

Time: 3 Hours Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Solve linear system of equations and to determine Eigen values and vectors of a matrix.
CO2: Exemplify the concept of vector space and subspace.
CO3: Determine Fourier series expansion of functions and transform.
CO4: Solve linear differential equation and integral equation using Laplace transform.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

(5 × 2 = 10)

			Marks	BL	CO	PO
1.	(a)	Find the characteristic equation of $A=\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix}$.	2	L1	1	2
	(b)	Show that $\{(1,0),(0,1)\}$ forms a basis for the vector space R^2 over R.	2	L1	2	2
	(c)	Is the set of all real numbers R over the set of all complex numbers C, a vector space? Justify your answer.	2	L2	2	2
	(d)	Show that $F_s\{x f(x)\} = -\frac{d}{ds}F_c\{f(x)\}$.	2	L1	3	2
	(e)	Find $L\{t \sin 5t\}$.	2	L2	4	

PART B

(4 × 10 = 40)

I.	(a)	Solve the system of equations $2x+3y+4z=31,\ x-2y+4z=4,\ 3x+y-3z=2,\ 6x+2y-6z=4$.	5	L3	1	
	(b)	Find the eigen values and eigen vectors of $A=\begin{bmatrix}1 & 0 & -1\\ 1 & 2 & 1\\ 2 & 2 & 3\end{bmatrix}$.	5	L4	1	

OR

	(a)	Find l and m if the rank of the matrix $A=\begin{pmatrix}0 & 1 & -3 & -1\\ 1 & 0 & 1 & 1\\ 3 & 1 & l & m\\ 1 & 1 & -2 & l\end{pmatrix}$ is 2.	5	L5	1	
	(b)	Using Cayley Hamilton theorem find the inverse of the matrix. $A=\begin{bmatrix}1 & 0 & 3\\ 2 & 1 & -1\\ 1 & -1 & 1\end{bmatrix}$	5	L4		

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			Marks	BL	CO
IV.	(a)	Determine whether the vectors (1, 3, 2), (1, -7, 8) and (2, 11, -1) are linearly independent.	5	L3	2
	(b)	Let V be the vector space of set of all 2×2 matrices. Prove that dim (V) = 4.	5	L4	2
		OR			
V.	(a)	Find an orthonormal basis for the subspace W of the vector space R^3 over the field R generated by the vectors (2, 1, 1) and (1, 3, -1).	5	L5	2
	(b)	Let V and W be two vector spaces over the same field F and let T be a linear transformation from V to W, prove that Ker T is a subspace of V and Im T is a subspace of W.	5	L4	2
VI.	(a)	Find the Fourier series expansion for $f(x) = \begin{cases} x + x^2, & -\pi < x < \pi \\ \pi^2, & x = \pm\pi \end{cases}$. Hence prove that $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \ldots\ldots$	5	L3	
	(b)	Find the Fourier integral representation of $f(x) = \begin{cases} 1, & \lvert x \rvert < 1 \\ 0, & \lvert x \rvert > 1 \end{cases}$. Hence prove that $\int_0^\infty \frac{\cos x \sin x}{x} dx \frac{\pi}{4}$.	5	L6	
		OR			
II.	(a)	Find the Fourier transform of $e^{-a^2x^2}$, $a > 0$, hence deduce that $e^{\frac{-x^2}{2}}$ is self reciprocal with respect to Fourier transform.	5	L5	
	(b)	Find the Fourier sine transform of $e^{-\lvert x \rvert}$. Hence evaluate $\int_0^\infty \frac{x \sin mx}{1 + x^2} dx$.	5	L6	
I.	(a)	Evaluate $\int_0^\infty \frac{e^{-4t} \sin^2 t}{t} dt$ using Laplace transform.	5	L	
	(b)	Find the inverse Laplace transform of $\frac{5s^2 - 15s - 11}{(s+1)(s-2)^3}$.	5	I	
		OR			
	(a)	State Convolution theorem and evaluate $L^{-1}\left\{\frac{1}{(s^2+1)(s^2+9)}\right\}$.	5		
	(b)	Solve $y'' - 3y' + 2y = e^{2x}$, $y(0) = -3$, $y'(0) = 5$.	5		

n's Taxonomy Levels

4.3%, L2 – 9.5%, L3 – 19%, L4 – 23.9%, L5 – 19%, L6 – 14.3%.

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